

Redefining the Human Stride

Cycloidal-Driven Assistive Hip Exoskeleton

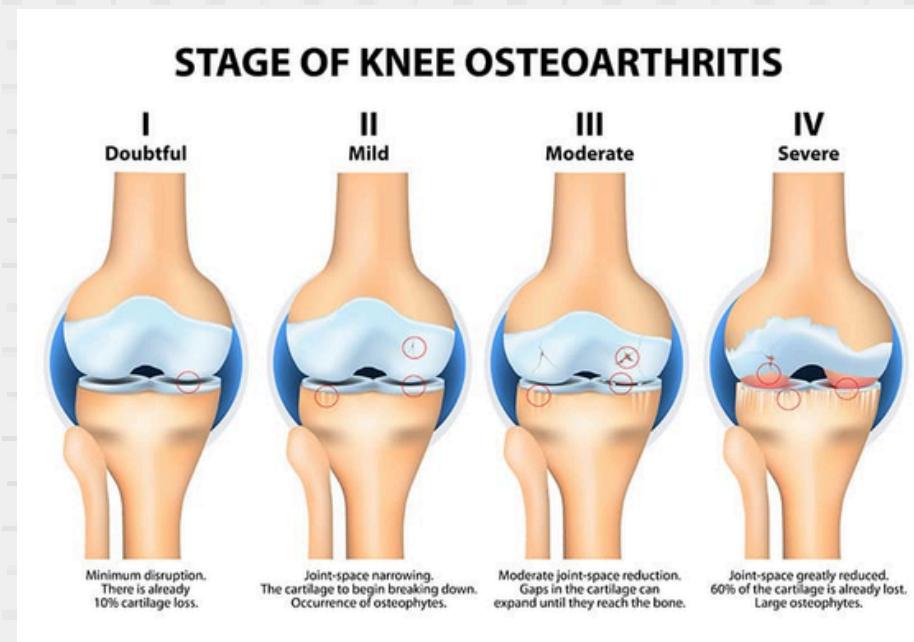
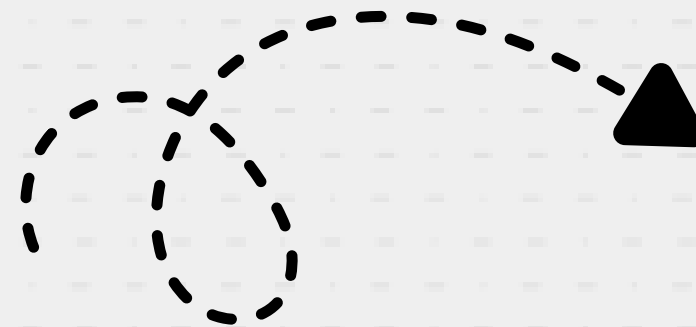
► PRESENTED BY:
YU CHENG YEE, LIU CHUANHUA, LEE JUN XIAN,
JACK LOW EE JIN, AND LIM CHU JUN

Problem Statement

The Freedom of Movement

How do we Aid Movement for the Ageing Population?

- **Provides lifting assistance** to make walking much easier.
- **Extends the time and distance** users can move without getting tired.

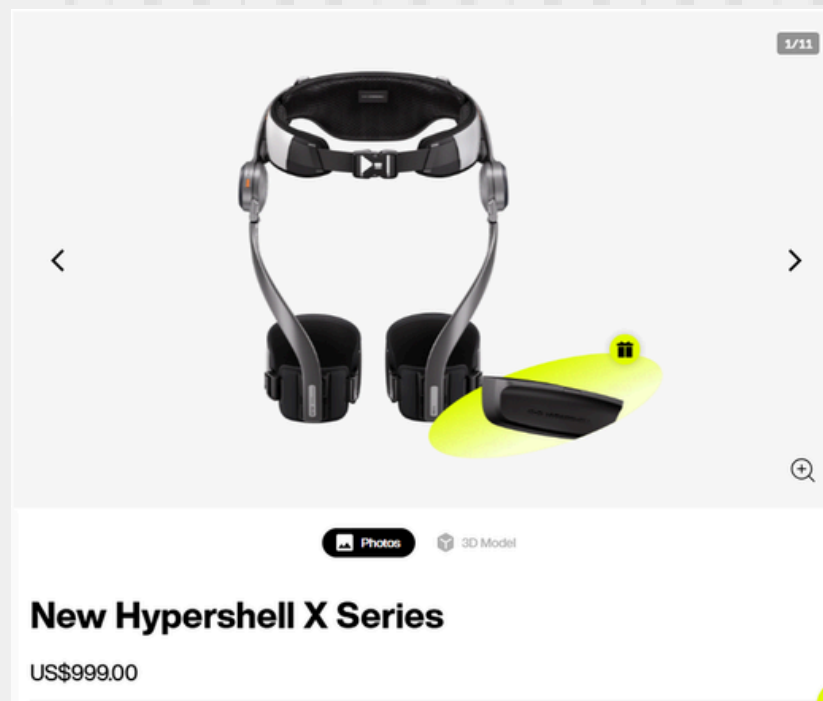


Exoskeleton !!

Problem Statement

However....

High Barrier to Entry.... **Expensive....**



US\$999.00

Huge & Clunky....



Our Vision

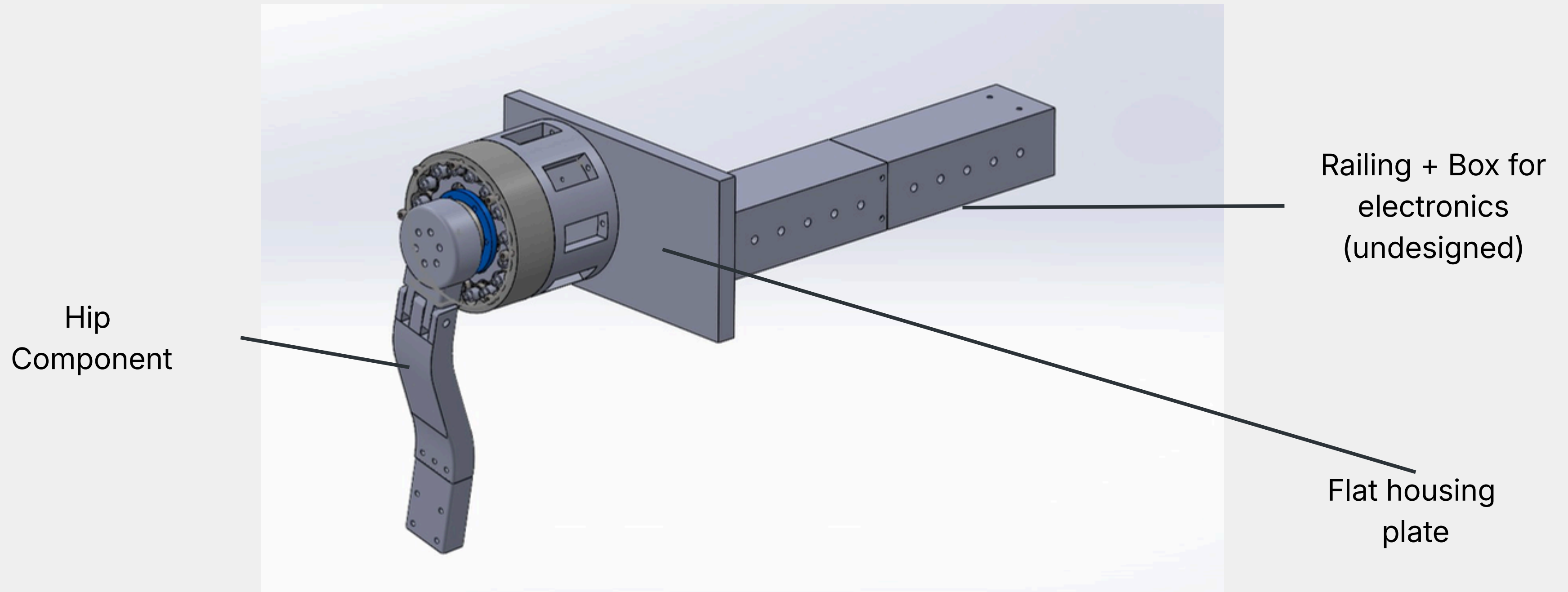
Create a...

lightweight
<4kg

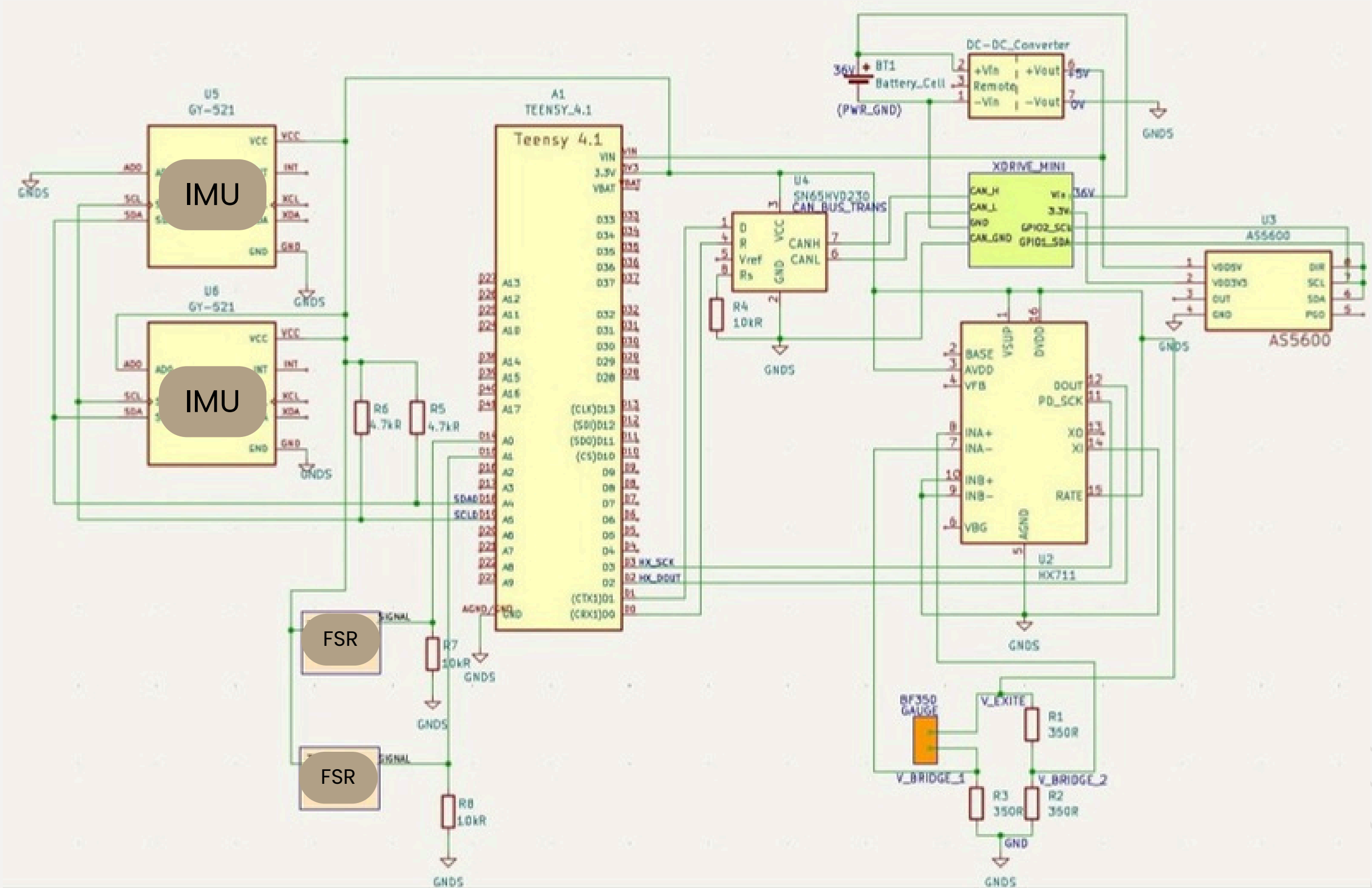
affordable
<\$1000

wearable device
supports daily mobility

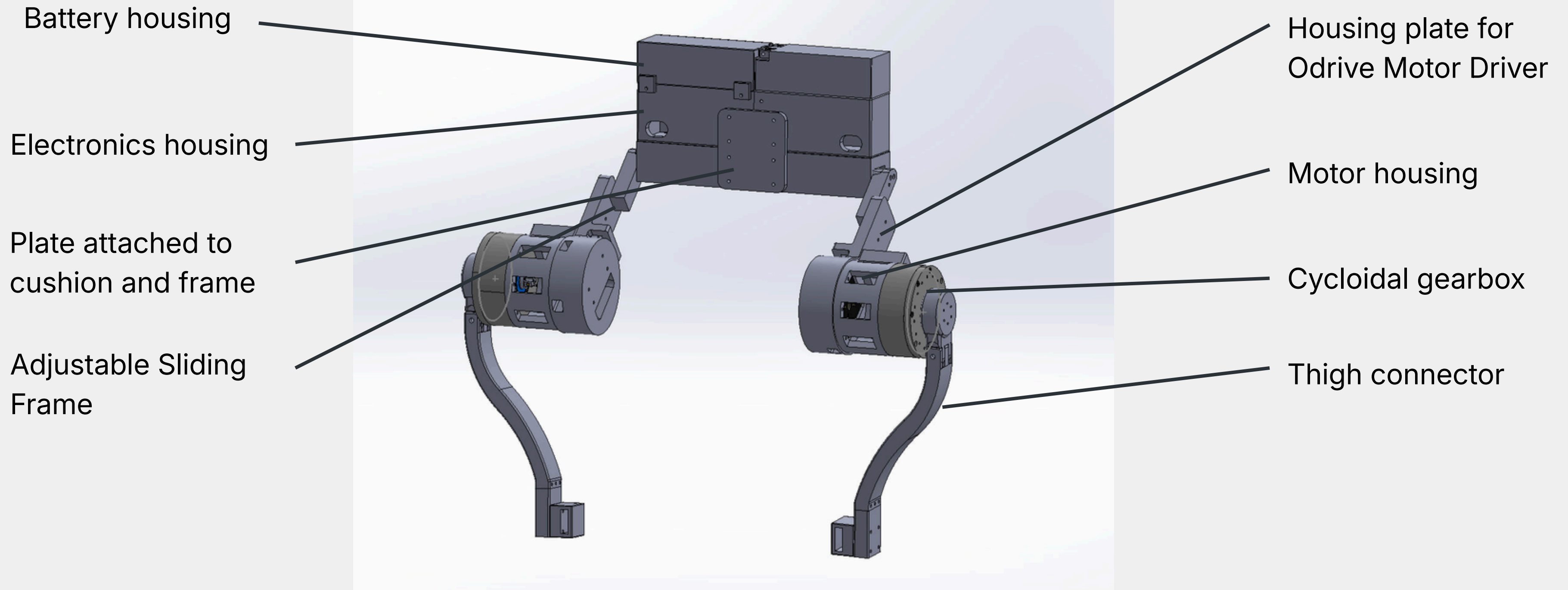
Proposed Initial Design



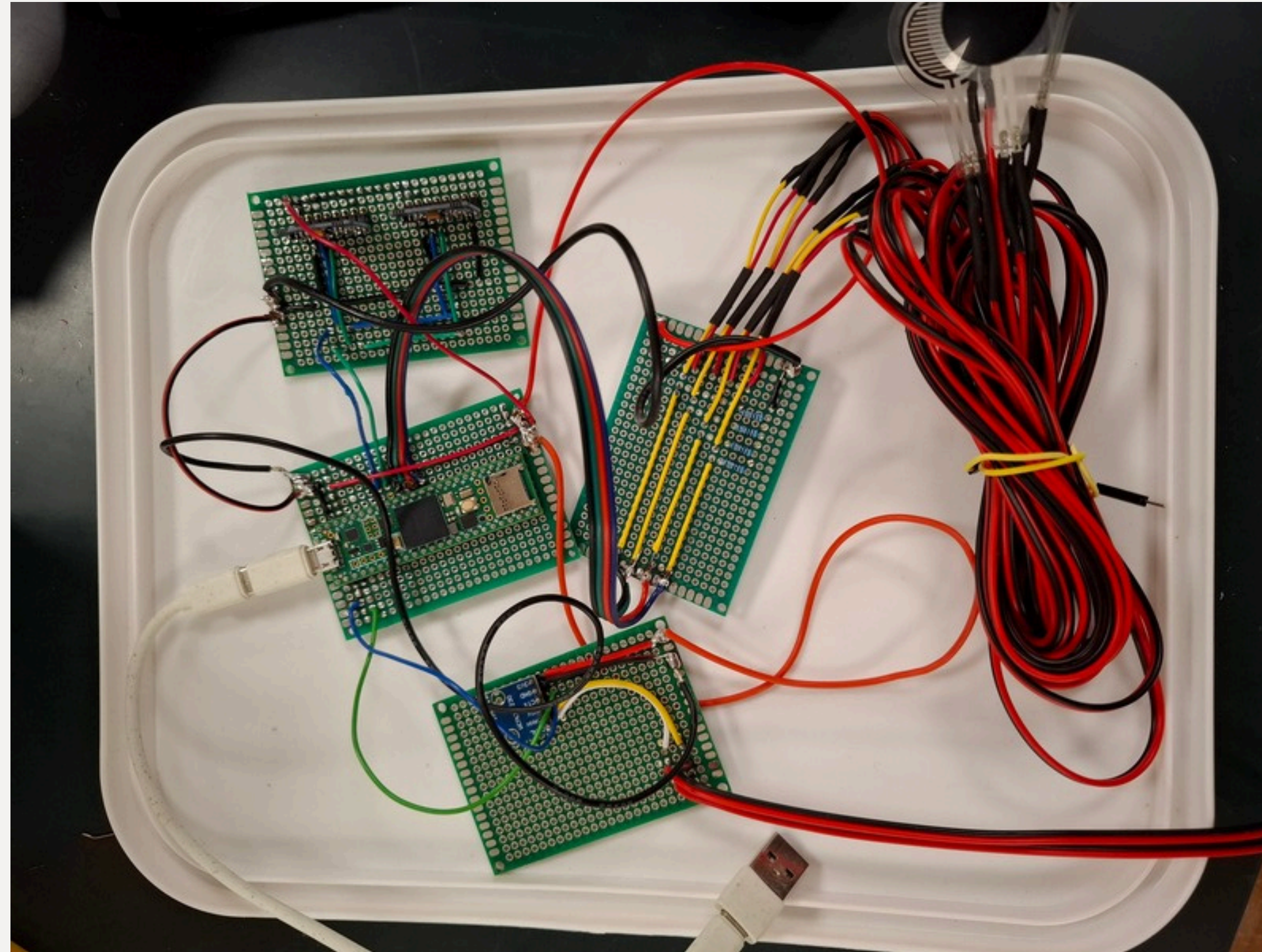
Initial Circuit Schematics



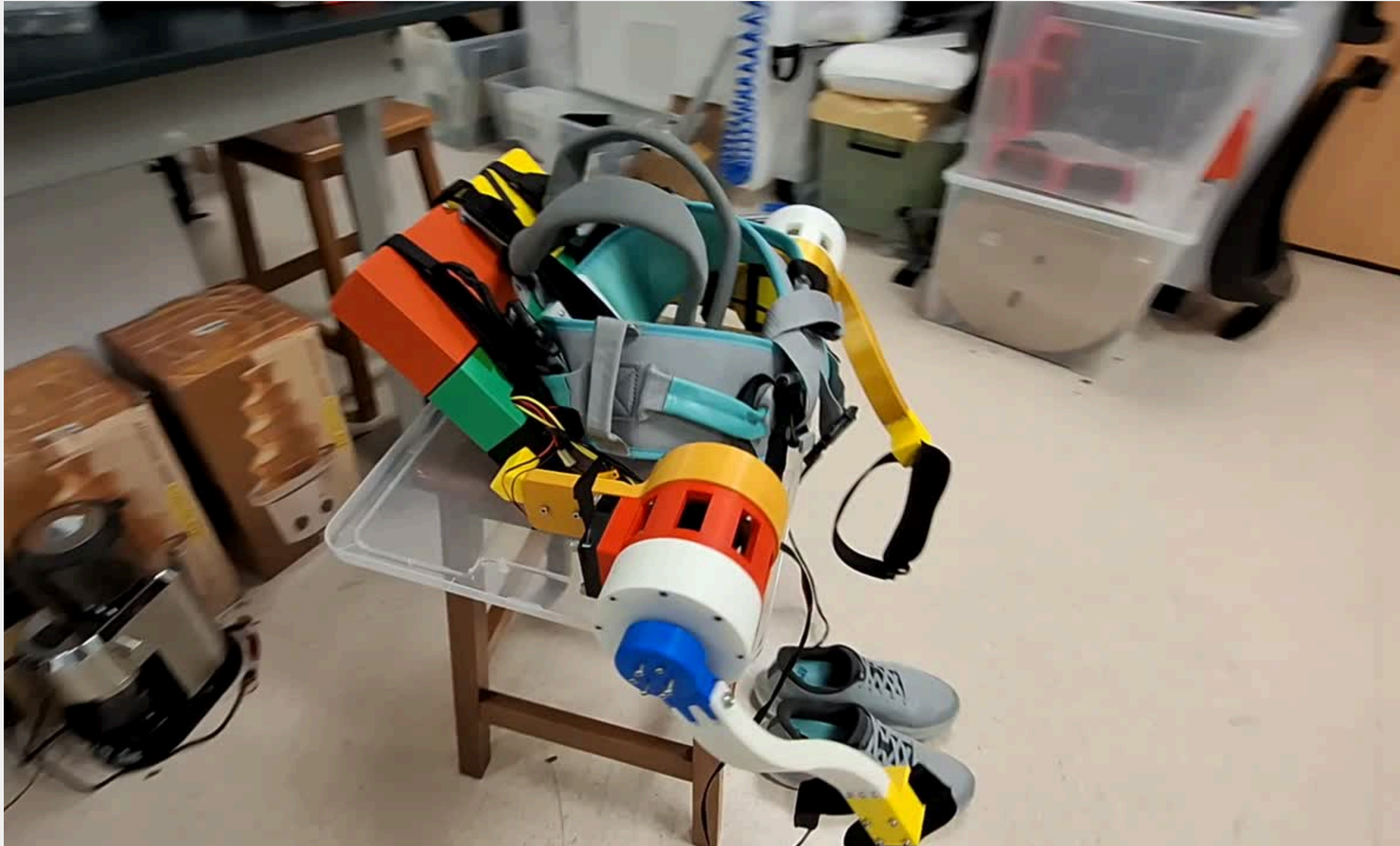
Proposed Final Design



Final Circuit Design/Circuitry



Final Product

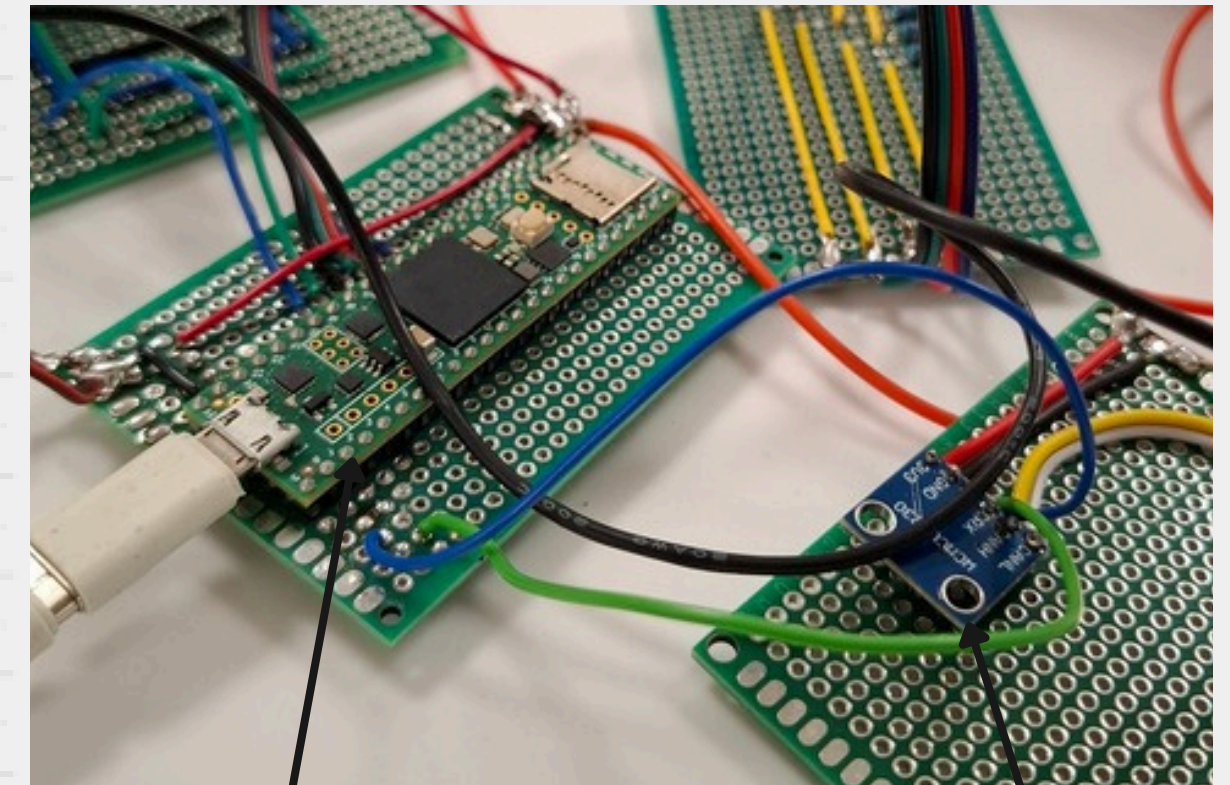


Overcame Challenges

Motor Contoller and MCU communication

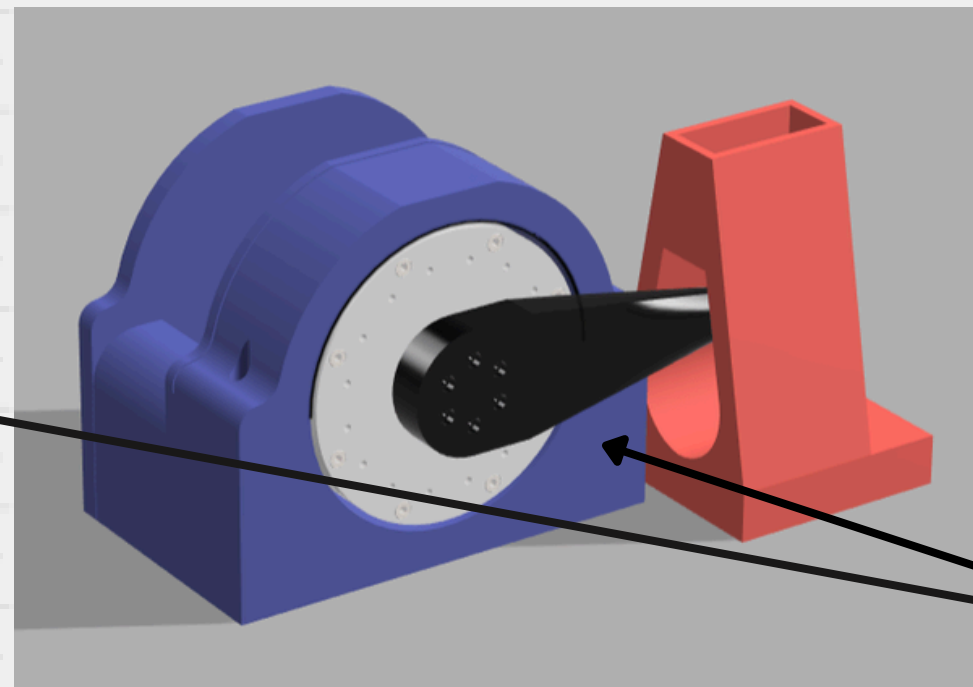
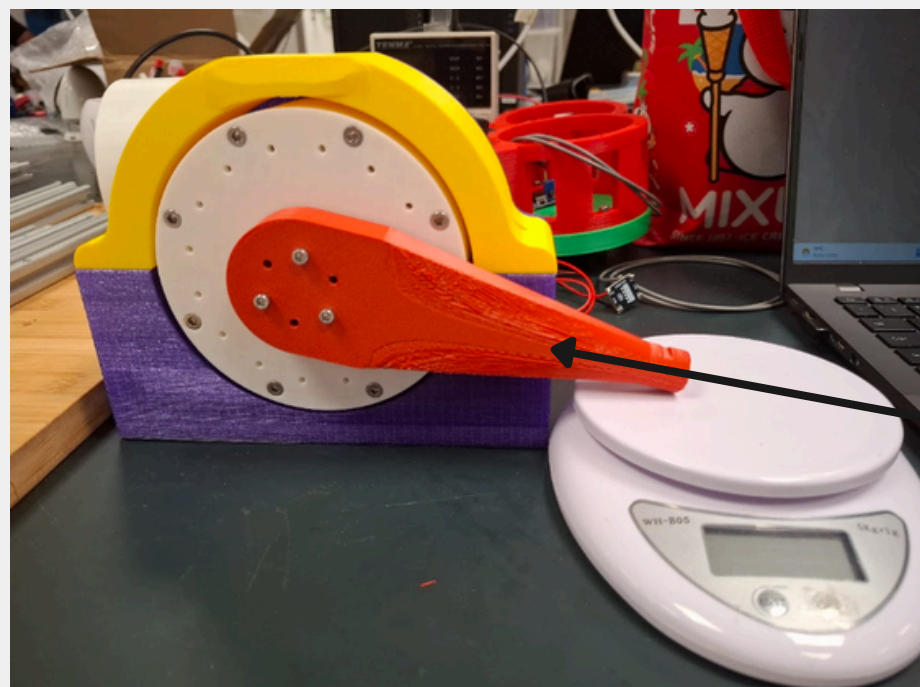
Problem: Code compatibility issue + CAN bus transceiver malfunctioning

Fix: Updated the code with suitable library + Replacement of the malfunctioning transceiver



Teensy 4.1 (MCU)

CAN Bus Transceiver



Incorrect Torque Output

Problem: Output torque differs from torque command

Fix: Updated firmware with torque tuning

Torque Testing Rig

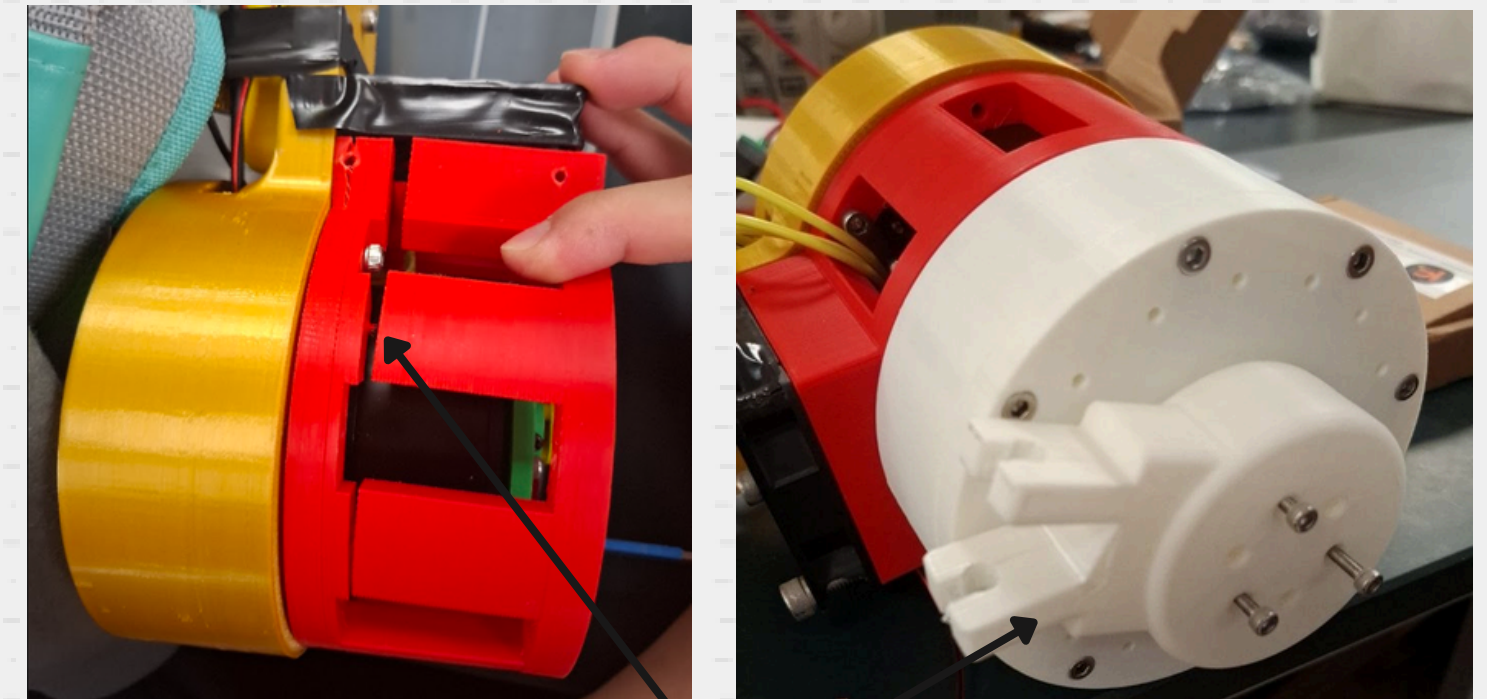


Overcame Challenges

Structural Failure Under Stress

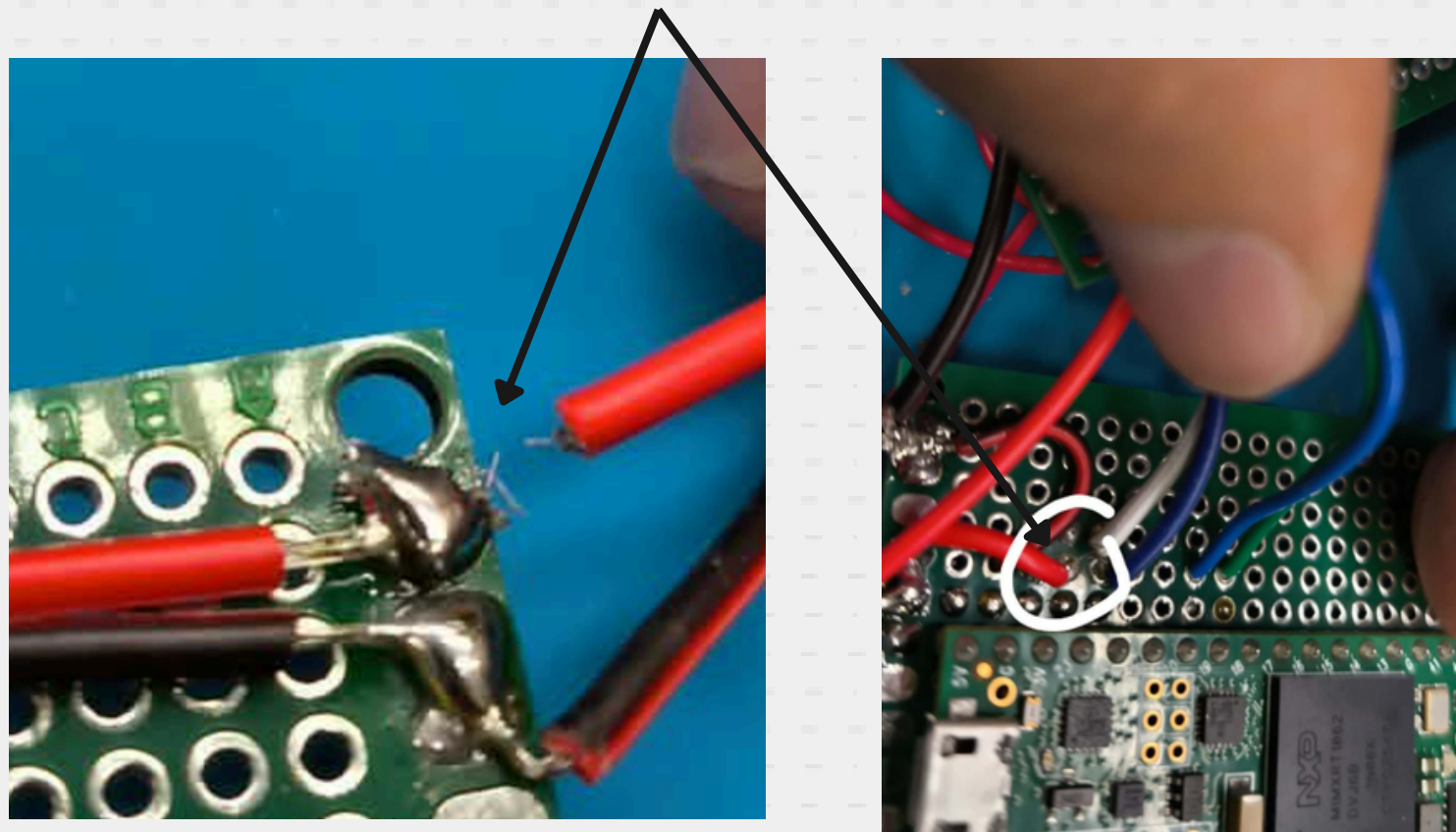
Problem: Mechanical failure and layer cracking under heavy actuation load

Fix: Added fillet radii to high-stress corners and increased shell thickness + infill density



Cracked under Heavy Load

Wires Snapped



Perfboard Wire Fatigue

Problem: Internal copper strands in soft wires repeatedly broke under movement at rigid solder connections causing long debugging time

Fix: Replace it to hard wires to eliminate wire flex at joint boundaries and use track soldering instead of wire

Challenges to Overcome

Weight and Size

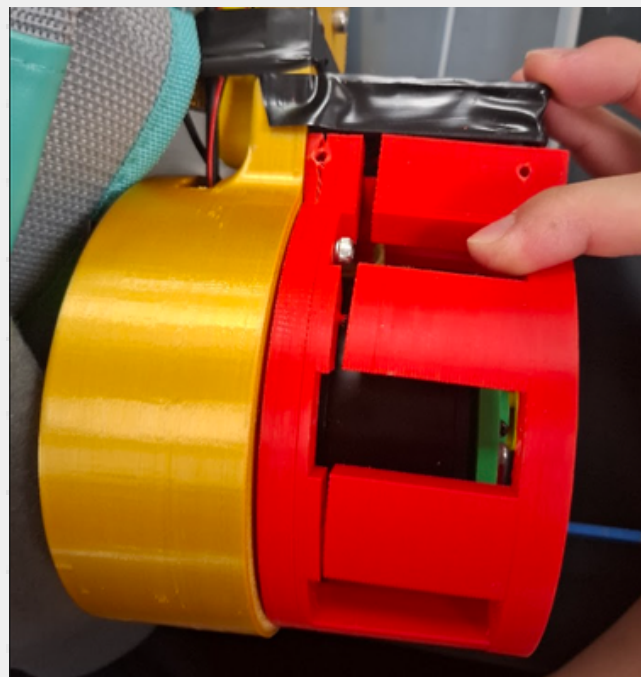
- Problem:
- Current weight is about 6kg
 - Clunkier than existing high-cost exoskeleton in the market (eg: Hypershell)
 - Requires customized inbuilt-sensor insole



Our Prototype



Hypershell



Structural Integrity

- Problem:
- Unable to support aggressive and prolonged movements
 - Current max torque: 5 Nm...

Bend away from the back
when applying high torque

Challenges to Overcome

Backend Logic

- Problem:
- Solely relies on the accuracy of the force sensor to trigger the actions of the motor.
 - Sometime misfire during runtime.
 - Cannot support multiple movement types (eg: running, jumping, hiking etc...)

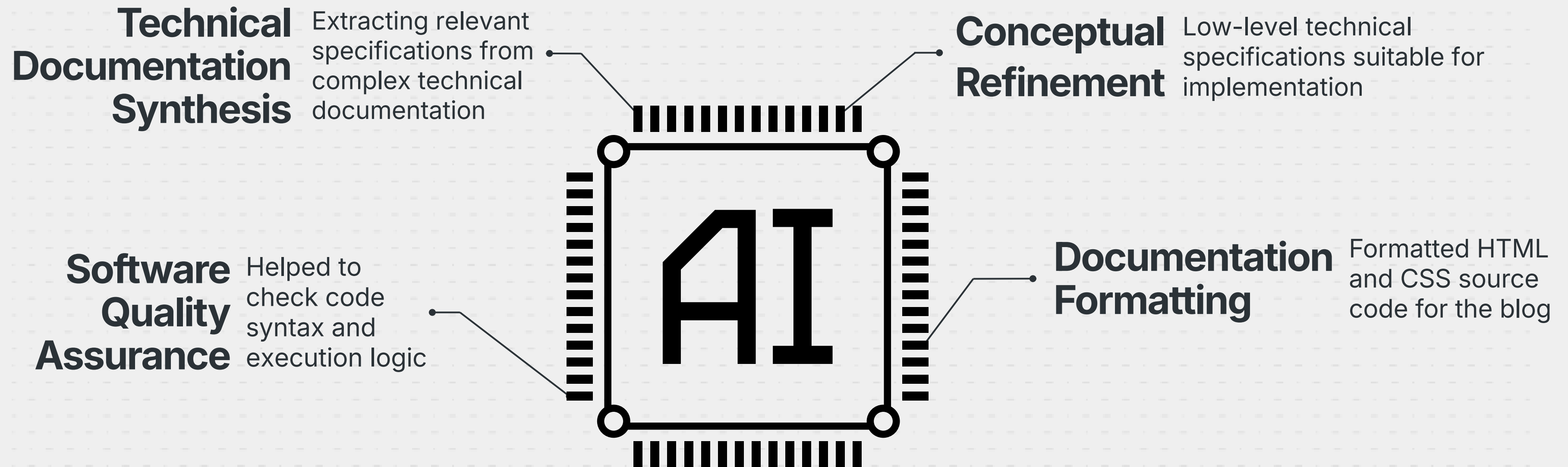


Film Force Sensor (FSR)



Different Movement Types

Role of AI



Future Plans

Structural Optimisation

- Finite Element Analysis and Topology Optimisation
- Lighter parts, better strength
- Using stronger materials (i.e. metal)

Custom Integrated Motor

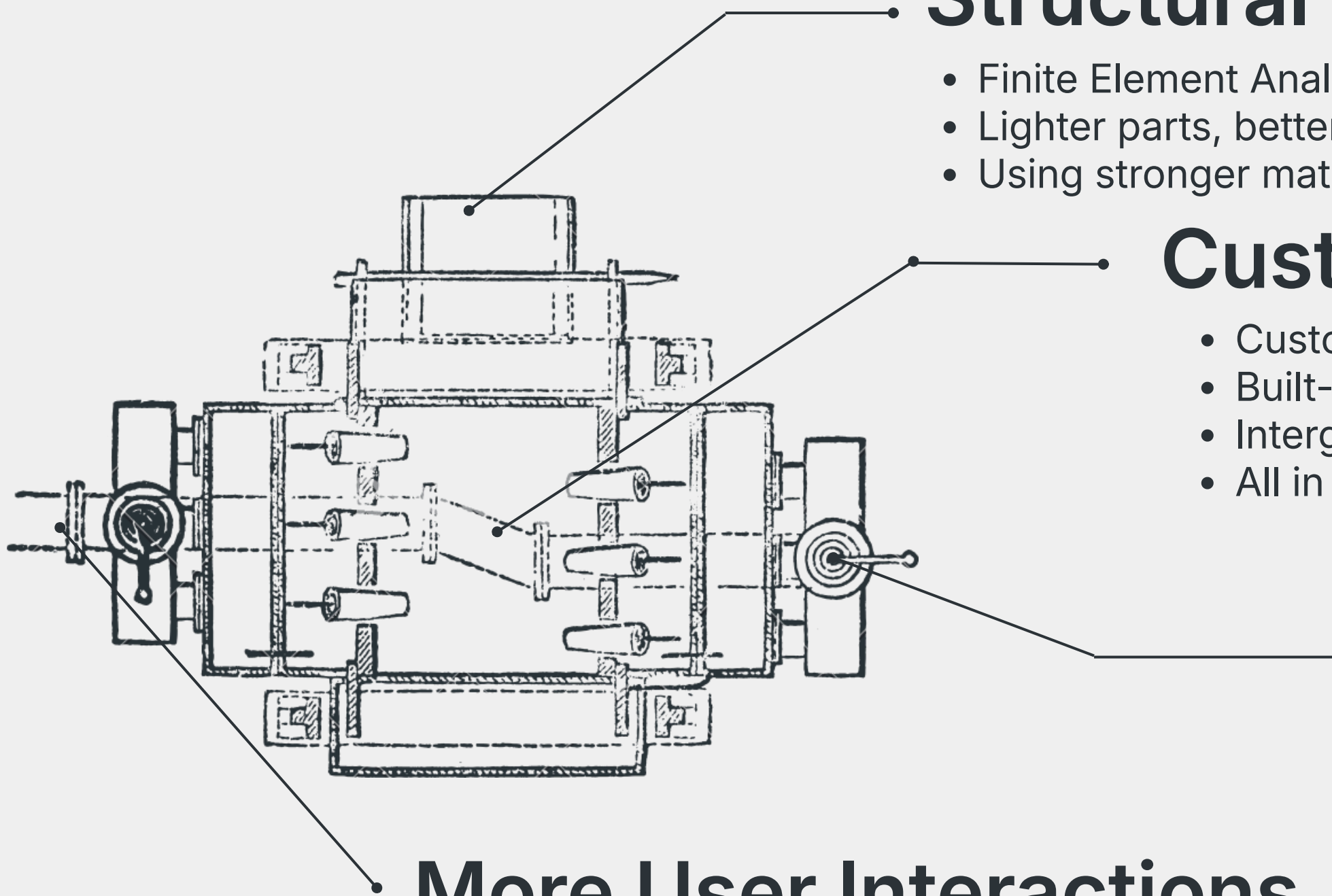
- Custom stator and rotor dimensions
- Built-in cycloidal gearbox
- Integrated cooling and encoder
- All in one actuation built for exoskeleton

Improved Logic & More Activity Types

- More comprehensive logic (eg: ML prediction)
- Expand usage to more movement types
 - Sitting
 - Walking Incline
 - Jumping
- More comprehensive daily mobility assistance

More User Interactions

- Develop an monitoring app
- Can switch modes, check battery life etc.



Final Expenditure

Total:
~\$ 840

Item	Option	Unit Cost	Quantity	Shipping & Tax	Total Item Cost	Remarks
Steadywin PM6025 Brushless Gimbal Motor	with AS5048A Encoder	\$104.37	2	\$22.25	\$230.99	
Makerbase MKS XDRIVE 56v Motor Controller	MKS XDRIVE MINI	\$44.31	2	\$3.99	\$92.61	
Film Pressure Sensor	100kg	\$1.92	4	\$6.66	\$14.34	
6 Axis IMU Gyroscope Accelerometer Sensor	bending needle	\$2.20	2	\$2.34	\$6.74	
Ball bearing 6x13x5mm 686-2RS		\$1.85	88	\$0.00	\$162.80	
Ball bearing 15x24x5 6802-2RS		\$1.72	8	\$0.00	\$13.76	
Ball bearing 35x47x7 6807-2RS		\$2.05	4	\$0.00	\$8.20	
Teensy 4.1 Development Board	with headers	\$66.20	1	\$0.00	\$66.20	
SN65HVD230 CAN bus transceiver communication-module for arduino		\$2.71	1	\$1.99	\$4.70	
304 Stainless Steel Cup Head Hexagon Socket	M3*30	\$6.75	1	\$1.99	\$8.74	Comes in 20 pcs/unit
Buzesty IT Band Strap		\$4.12	4	\$1.99	\$18.47	
Gait Belt with Straps	L size	\$24.88	1	\$0.00	\$24.88	
Reusable Fastening Cable Straps	20cm 30 Pcs	\$5.70	1	\$1.99	\$7.69	
Cooling Fans	5V	\$3.01	2	\$1.99	\$8.01	
36V Lithium Battery Pack	15Ah DC-XT60	\$67.98	1	\$6.12	\$74.10	
Socket Hexagon Cap Head Bolts 304 Stainless	M4*25, M4*30	\$0.40	60	\$1.99	\$25.99	
Self Tapping Screws	ST4.2, 45mm	\$2.16	2	\$1.99	\$6.31	Comes in 10 pcs/unit
Lead-Free Needle Solder Paste		\$4.02	1	\$1.99	\$6.01	
36V Lithium Battery Charger		\$6.49	1	\$3.47	\$9.96	
Solder Breadboard Double Sided 50mm x 70mm		\$2.00	4	\$0.00	\$8.00	
DC-DC Step Down Buck Converter		\$1.69	2	\$3.50	\$6.88	
Wire Splitter	1 input 3 output	\$2.77	1	\$3.14	\$5.91	Comes in 5 pcs/unit
Arch support Sports insoles	L size EU41-46)	\$4.99	1	\$1.99	\$6.98	
7A Switch Fuse Resettable 220V		\$8.51	1	\$1.99	\$10.50	
240V AC 4-Pin Rocker On Off Push Button Switch		\$8.61	1	\$1.99	\$10.60	
			Total		\$839.37	

*BoM is only based on the finalised materials bought AND used to assemble the full exoskeleton. Full BoM is slightly above \$1k.

Demo Video



PS5888
Making and Tinkering





PS5888
Making and Tinkering

Thank You!

Any Questions?

